INF20325 Estructuras De Datos

Ayudantía 8

Departamento de Informática
Universidad Técnica Federico Santa María
13 de Octubre de 2026

Ejercicio 1: Inserte en un árbol 2-3 (inicialmente vacío) la siguiente secuencia de números:

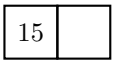
{15, 23, 10, 7, 12, 18, 21, 27, 25, 30, 6, 3}

¿Cuál es la **mínima** cantidad de elementos que se deberían insertar al árbol 2-3 resultante para que sea **equivalente a un ABB**? *Ejercicio similar a C2 2023.1.*

Solución:

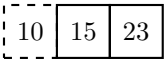
1)

{~~15~~, 23, 10, 7, 12, 18, 21, 27, 25, 30, 6, 3}



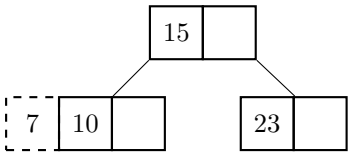
2)

{~~15, 23, 10~~, 7, 12, 18, 21, 27, 25, 30, 6, 3}



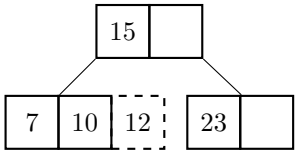
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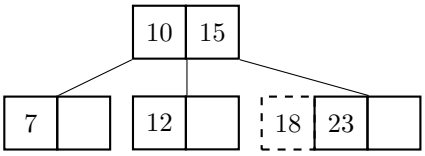
6)

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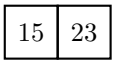
8)

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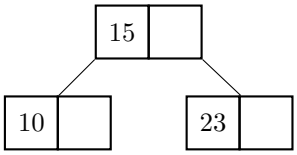
2)

{~~15, 23,~~ 10, 7, 12, 18, 21, 27, 25, 30, 6, 3}



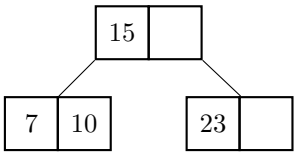
3)

{~~15, 23, 10~~, 7, 12, 18, 21, 27, 25, 30, 6, 3}



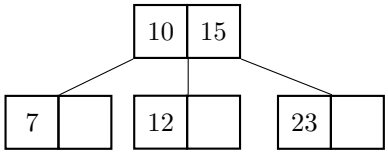
5)

{~~15, 23, 10, 7~~, 12, 18, 21, 27, 25, 30, 6, 3}



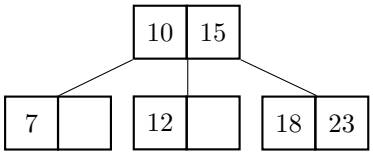
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{~~15, 23, 10, 7, 12,~~ 18, 21, 27, 25, 30, 6, 3}



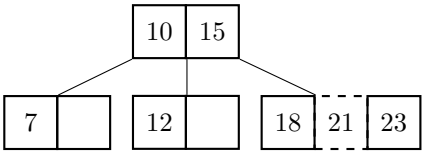
9)

{~~15, 23, 10, 7, 12, 18,~~ 21, 27, 25, 30, 6, 3}



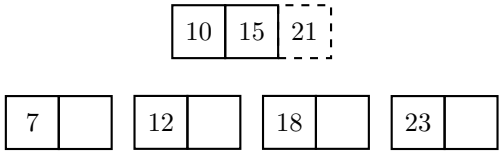
10)

{~~15, 23, 10, 7, 12, 18, 21,~~ 27, 25, 30, 6, 3}



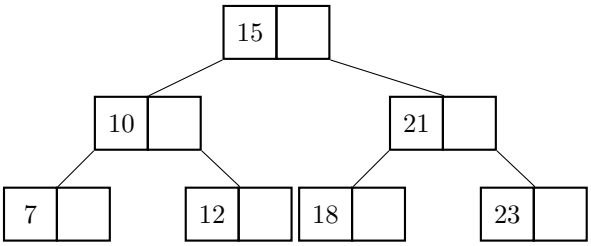
11)

{~~15, 23, 10, 7, 12, 18, 21,~~ 27, 25, 30, 6, 3}



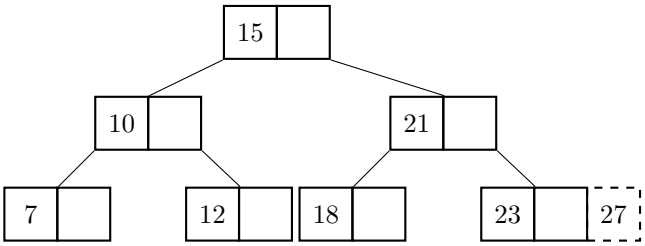
12)

{~~15, 23, 10, 7, 12, 18, 21,~~ 27, 25, 30, 6, 3}



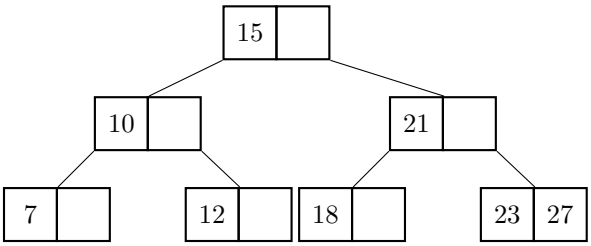
13)

{~~15, 23, 10, 7, 12, 18, 21, 27,~~ 25, 30, 6, 3}



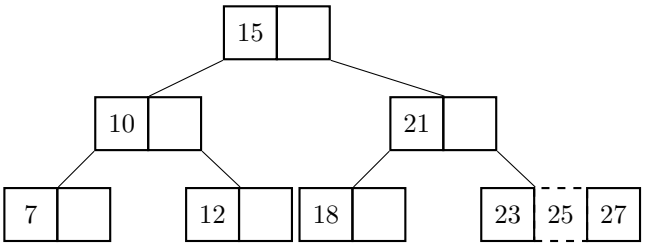
14)

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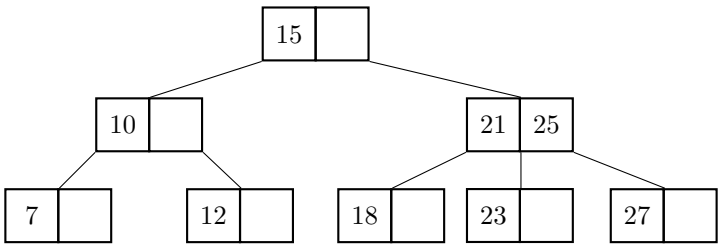
15)

{~~15, 23, 10, 7, 12, 18, 21, 27, 25,~~ 30, 6, 3}



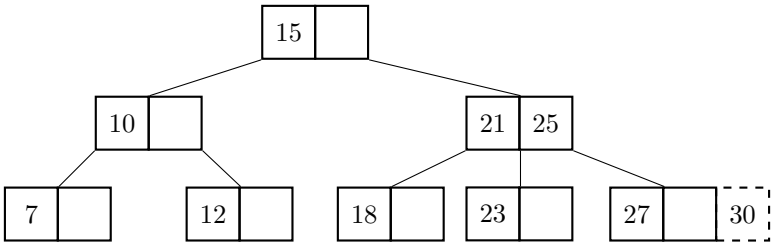
16)

{~~15, 23, 10, 7, 12, 18, 21, 27, 25,~~ 30, 6, 3}



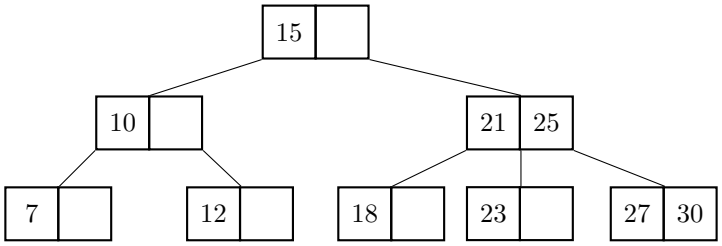
17)

{~~15, 23, 10, 7, 12, 18, 21, 27, 25, 30,~~ 6, 3}



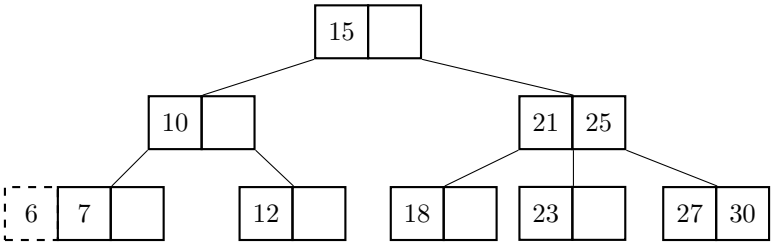
18)

{~~15, 23, 10, 7, 12, 18, 21, 27, 25, 30,~~ 6, 3}



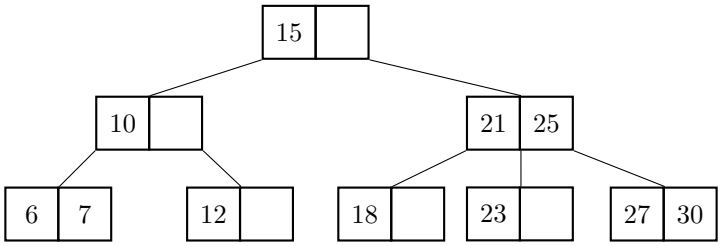
19)

{~~15, 23, 10, 7, 12, 18, 21, 27, 25, 30, 6,~~ 3}



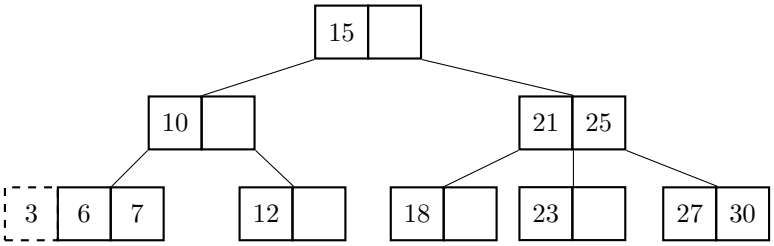
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{~~15, 23, 10, 7, 12, 18, 21, 27, 25, 30, 6,~~ 3}



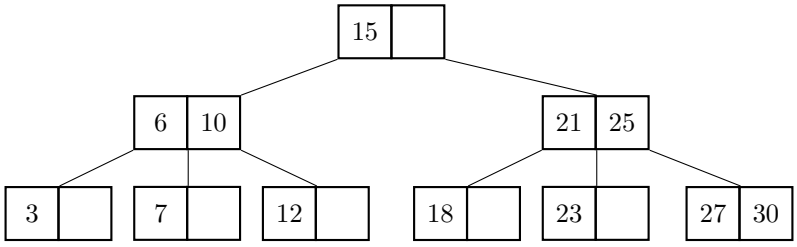
21)

{~~15, 23, 10, 7, 12, 18, 21, 27, 25, 30, 6, 3~~}



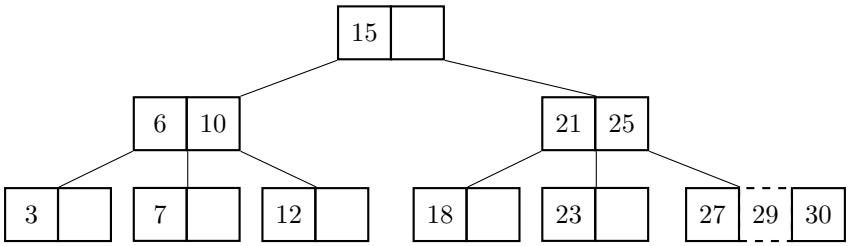
22)

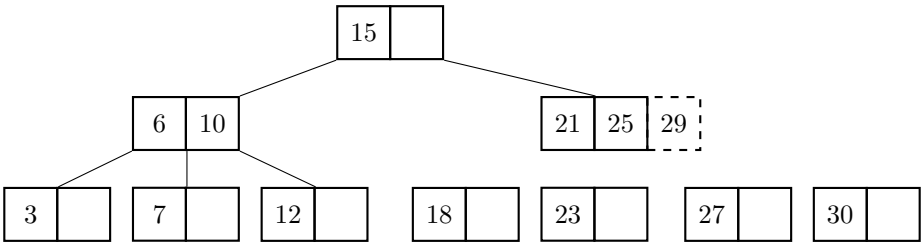
{~~15, 23, 10, 7, 12, 18, 21, 27, 25, 30, 6, 3~~}

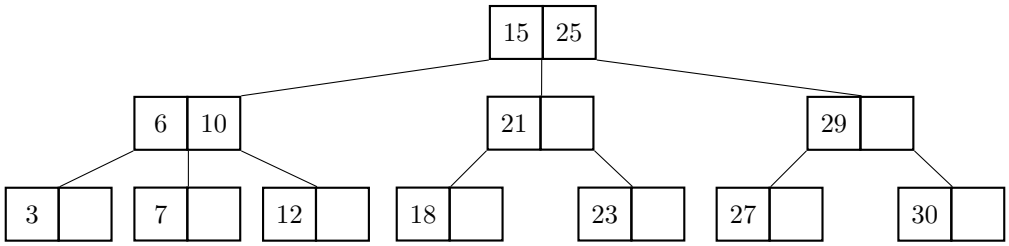


Para transofrmar este árbol 2-3 en uno equivalente a un ABB, es necesario que **cada nodo tenga un solo elemento, y exactamente 2 hijos** *(como pequeño recordatorio, árboles 2-3 no pueden tener 1 hijo).*

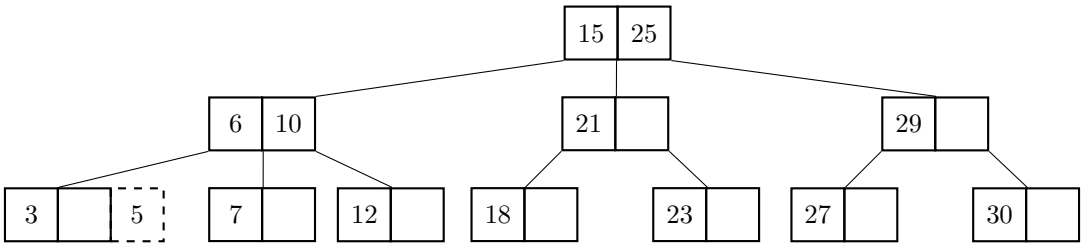
Podemos deshacernos de los tres hijos de (21,25) insertando un elemento cuyo valor sea **mayor a 25**. En este caso, se usará el valor *29*, pero puede ser cualquiera en ese rango:

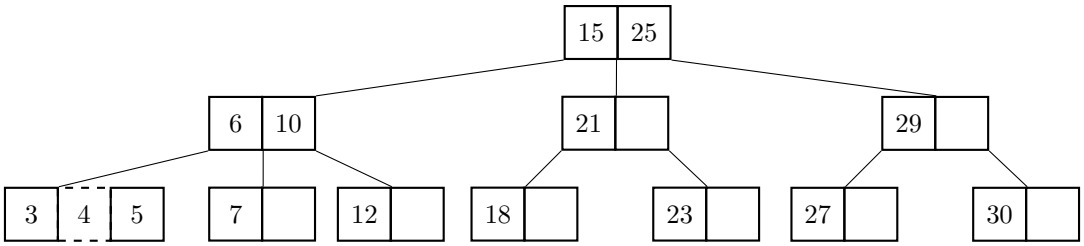


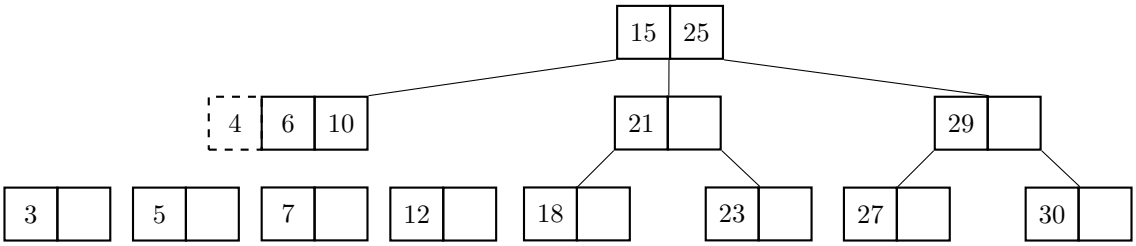


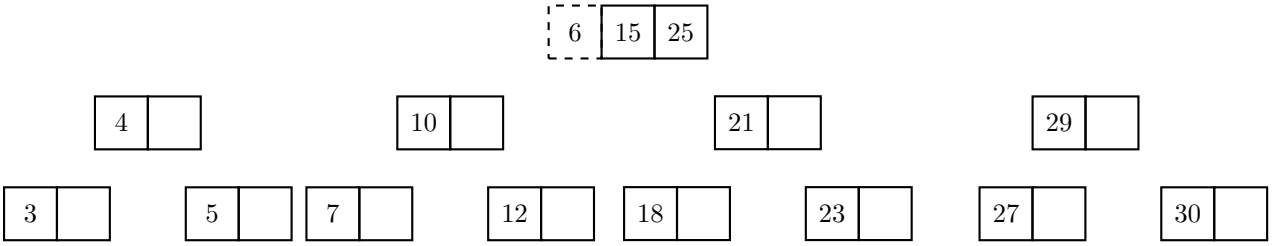


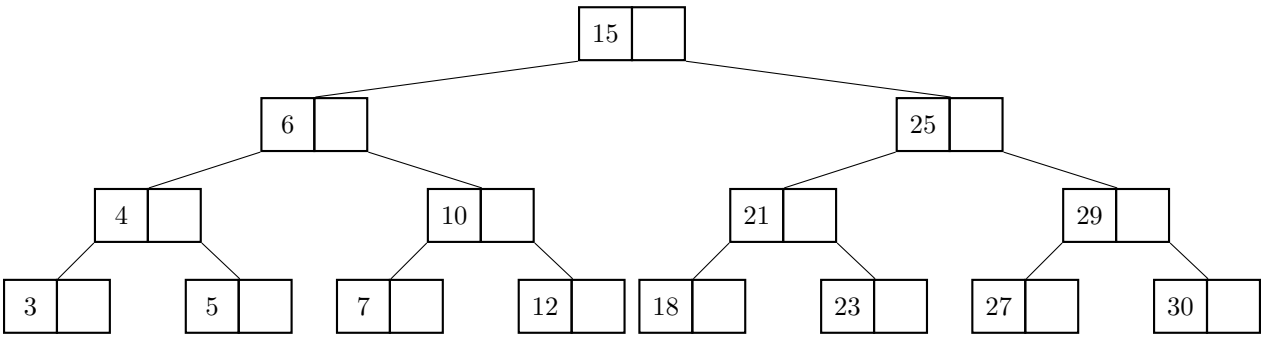
Ahora, tenemos que realizar el mismo proceso en el subárbol izquierdo (mirándolo desde la raíz). Para ello, insertaremos dos elementos con valor **menor a 6, pero mayor a 3**. En este caso, insertaremos 5 y luego 4:




15 25
6 10
21
29
3 4 5
7
12
18
23
27
30


15 25
4 6 10
21
29
3
5
7
12
18
23
27
30


6 15 25
4
10
21
29
3
5
7
12
18
23
27
30


15
6
25
4
10
21
29
3
5
7
12
18
23
27
30

Finally...

Ejercicio 2: Implemente una función que reciba un string con código de *Scheme*, un lenguaje cuya syntax está basada en paréntesis, que retorne una **lista con las posiciones de inicio y fin de cada set de paréntesis**. En caso de haber un error de paréntesis mal cerrados, se debe retornar una **lista vacía**. Suponga que tiene a su disposición el struct **Rango**, que encapsula dos variables enteras inicio y fin.

Por ejemplo, el siguiente string:

(* (- 8 7) (+ 4 6))

Retorna una lista de rangos:

(0,18), (3,9), (11, 17)

Solución:

```
tLista funciones(char* codigo){
    tPila p;
    tLista res;
    tLista error;
    int pos = 0;

    while(codigo[pos] != '\0'){
        if(codigo[pos] == '('){
            p.push(pos);
        } else if (codigo[pos] == ')'){
            if (p.size() == 0) return error;

            Rango r;
            r.inicio = p.topValue();
            r.fin = pos;
            res.append(r);
            p.pop();
        }
        pos++;
    }

    if (p.size() > 0) return error;
    return res;
}
```